

Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____

Factoring the GCF

Example (Factor out the GCF):

$6x^2 + 9x$ — the GCF is $3x$. Divide each term: $6x^2/3x = 2x$ and $9x/3x = 3$. So $6x^2 + 9x = 3x(2x + 3)$.

Directions: Factor out the GCF. Check by distributing.

1. $4x + 8$

4. $12x^2 - 18x$

2. $10x - 15$

5. $4x^3 - 8x^2$

3. $6x^2 + 9x$

Challenge Problems

6. $15x^3 + 25x^2 - 10x$ — factor out the GCF completely.

Answer Key and Procedure

Procedure

1. Find the GCF of the coefficients.
2. Take the smallest power of each shared variable.
3. Write GCF times the leftover factor; distribute to check.

Answers

1. $4(x + 2)$
2. $5(2x - 3)$
3. $3x(2x + 3)$
4. $6x(2x - 3)$
5. $4x^2(x - 2)$
6. $5x(3x^2 + 5x - 2)$